

Dr Kostas Vasilopoulos

AI/ML Engineer @ Pfizer | Econ (Ph.D.)

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SUMMARY

Machine Learning Engineer and Ph.D. in Economics building enterprise ML and GenAI platforms in regulated environments. Lead engineering for Pfizer's **Data Science & GenAI platform (Dataiku)**, serving **1,800+ data scientists** and **1M+ LLM requests/month** across LLM integration, RAG, LLMOps, and platform standards for security and lifecycle management. An economics background brings rigor to decisions under uncertainty: model risk, tradeoffs, and what “good enough” means at scale.

EXPERIENCE

Senior Manager, Data Science Platform Engineering

Pfizer

📅 06.2026 – Present 📍 Thessaloniki, Greece

- Own the engineering strategy and roadmap for Pfizer's enterprise, cloud-native **Data Science & GenAI platform (Dataiku)**, serving **1,800+ data scientists** and analysts and hosting a large portfolio of production Python services and web apps.
- Scale **LLMOps** for **1M+ LLM requests/month**, driving cost controls, observability, and access governance across production ML and GenAI workloads.
- Architect enterprise **GenAI** capabilities (**LLM integrations, RAG** systems, AI assistants) as reusable platform services that accelerate AI adoption across the organization.
- Establish engineering standards, reusable frameworks, **APIs**, and **self-service tooling** that improve developer experience and model lifecycle governance in regulated environments.
- Provide **technical leadership** across platform, security, data science, and business teams, and manage a direct report.

Machine Learning Engineer, Manager

Pfizer

📅 05.2023 – 05.2026 📍 Thessaloniki, Greece

- Technical owner for integrating and scaling **Large Language Models (LLMs)** on the enterprise Dataiku platform, solving reliability, performance, and compliance constraints hands-on.
- Designed **MLOps** execution patterns for model deployment, monitoring, and access control in production, later adopted as **platform-wide standards**.
- Built reusable platform abstractions and reference implementations for Python services and web apps, driving **cross-team adoption** of AI capabilities.
- **Technical Lead for Ethicara ML**: designed and delivered responsible AI solutions aligned with enterprise and regulatory standards.
- Governed platform usage, onboarding, and operations across **users, administrators, and engineering teams**.

Machine Learning Engineer

OrosDynamics (formerly SophoTree)

📅 01.2022 – 05.2023 📍 United Kingdom

Promoted from Data Scientist (06.2021 – 01.2022)

EDUCATION

Ph.D. in Economics

Lancaster University

📅 2016 – 2020 📍 Lancaster, UK

Published research in peer-reviewed journals. Presented work at multiple international conferences and academic workshops.

M.Sc. in Economics, Applied Finance

University of Macedonia

📅 10.2014 – 03.2016 📍 Thessaloniki, Greece

B.Sc. in Economics

University of Macedonia

📅 10.2009 – 11.2014 📍 Thessaloniki, Greece

AI & MACHINE LEARNING

- Generative AI & Large Language Models (**RAG, embeddings, fine-tuning**)
- Natural Language Processing & Text Analytics
- Production ML Operations & Model Lifecycle
- Model Evaluation, Monitoring & Drift Detection

ENGINEERING & INFRA

- Programming: **Python, R, SQL, JavaScript**
- Distributed Computing & Microservices Architecture
- Cloud & Containers: **AWS, Azure, Docker, Kubernetes**
- Scalability & Performance Optimization
- Web Development & **REST APIs**
- Data Engineering: **ElasticSearch, Postgres, Kafka, NoSQL**
- DevOps: **Git, Linux, CI/CD Pipelines**

- **Tech Lead** for EU Horizon 2020 initiatives (**Infinitech**, **AI4PP**), owning end-to-end system architecture, delivery, and long-term technical direction.
- Designed and implemented a **distributed data collection and processing platform** on Kubernetes, horizontally scaling to **millions of jobs per day**.
- Productionized multiple ML models with autoscaling, microbatching, and runtime optimizations, increasing throughput by **320%** and cutting infrastructure costs by **40%**.
- As de facto **platform owner**, defined reusable ML pipelines and **CI/CD**, observability, and reliability standards adopted across models, services, and teams.
- Built and operated API microservices integrating **PostgreSQL**, **Elasticsearch**, and analytics pipelines for real-time and batch workloads.
- Led development and deployment of production NLP pipelines using **Spark NLP** (classification, sentiment analysis, NER), later reused across document and news processing workloads.

Data Scientist

OrosDynamics (formerly SophoTree)

📅 06.2021 – 01.2022 📍 United Kingdom

- Developed and trained NLP models for large-scale news and document analysis, focusing on **classification**, **sentiment analysis**, and **entity extraction**.
- Extended the platform from structured data to **unstructured text**, laying the groundwork for the production NLP pipelines later scaled as ML Engineer.

Visiting Researcher

Lancaster University

📅 07.2021 – 07.2022 📍 Lancaster, UK

- Designed and implemented real-time **exuberance detection and predictive modeling systems**, translating advanced econometric theory into production-grade open-source software (**exuber**).
- Delivered **14–150× performance improvements** via optimized matrix inversion and numerical methods, enabling real-time processing of large-scale economic datasets.
- Work from the Housing Observatory featured in **The Times** and used by practitioners, including **central banks**.

Teaching Associate

Lancaster University

📅 11.2020 – 08.2021 📍 Lancaster, UK

- Taught undergraduate and postgraduate economics courses, consistently receiving outstanding student evaluations.

Research Assistant & Graduate Teaching Assistant

Housing Observatory, Lancaster University

📅 10.2016 – 10.2020 📍 Lancaster, UK

- Built analytical models and interactive tools for real-time monitoring of UK national and regional housing markets.
- Led web development of the Housing Observatory platform (<https://housing-observatory.com/>), collaborating with the **Federal Reserve Bank of Dallas**.
- Authored and maintained 6 open-source R packages (**exuber**, **ivx**, **uklr**, **onsr**, **transx**, **ihpdr**) with **100K+ collective downloads**.

SELECTED PUBLICATIONS

- 📄 Vasilopoulos, K., & Tayler, W. (2026). Real Estate and Construction Sector Dynamics over the Business Cycle. *Economica*. doi:10.1111/ecca.70029
- 📄 Vasilopoulos, K., Pavlidis, E., & Martínez-García, E. (2022). Exuber: Recursive Right-Tailed Unit Root Testing with R. *Journal of Statistical Software*, 103, 1-26.
- 📄 Pavlidis, E.G., & Vasilopoulos, K. (2020). Speculative Bubbles in Segmented Markets: Evidence from Chinese Cross-Listed Stocks. *Journal of International Money and Finance*, 109, 102222.

SOFTWARE CONTRIBUTIONS

- </> **exuber**: Econometric Analysis of Explosive Time Series. Published in *Journal of Statistical Software* (2022).
- </> **ivx**: Robust Econometric Inference
- </> **uklr**: Client to United Kingdom Land Registry
- </> **onsr**: Client for the ONS API
- </> **transx**: Transform Univariate Time Series
- </> **ihpdr**: Download Data from the International House Price Database

📄 Collective downloads exceed 100,000. The complete package list is available at: [r-universe](#)

AWARDS & ACHIEVEMENTS

- ★ Departmental Studentship Award
Lancaster University, 2016–2019
- ★ MSc Scholarship
Awarded based on academic excellence, University of Macedonia